

Contact Resistance Test (Scope CRM 100B)

→ To check how healthy the breaker
Main Contact are

high resistance → bad result → Current flow
path is not
healthy



Heat happens
(I^2R)



CB failure

Low Resistance → good result → Current flow
(Contacts are clean) path is healthy.

Injected DC Current → 100A

Injected DC Voltage → 1-10V

Resistance is Shown in CRM

→ Resistance is Calculated using ohm's law with Voltage and Current Value

$$R = \frac{V}{I}$$

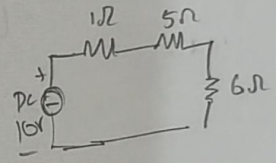
$$I \rightarrow 100A$$

$V \rightarrow$ Voltage drop at contact resistance (2.4mV)

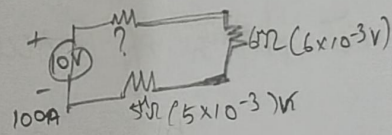
$$R = \frac{2.4 \times 10^{-3}}{100} = \frac{24 \times 10^{-4}}{100}$$

$$R = \frac{0.0024}{100}$$

$$R = 0.000024 = \boxed{24 \mu\Omega}$$



$$I = \frac{10}{12}$$



$$100 \times (R_1 + 5 + 6) = 10$$

$$\frac{0.03}{10^{-2}}$$

Insulation Resistance Test (RISH Insu 5D+)

- To check how healthy the insulation is between
- * phase to earth
 - * phase to phase

[No Leakage Current → good insulation



high resistance
(Giga ohms)]

- Health of insulation b/w
- phase and earth
 - phase and phase

→ Calculations

$$R = \frac{V}{I} = \frac{5000 \text{ V}}{0.0167 \mu\text{A}}$$

$$R = 300 \text{ G}\Omega$$

Injected DC Voltage $\rightarrow 5000V$

Why dc Voltage?

[If AC is used $\rightarrow$ Reactance $\rightarrow$ This impedance]
 $(X_L) = 2\pi fL$

$\rightarrow$ we need only resistance

DC $\rightarrow$ freq = 0 Hz $\rightarrow$ Inductive Reactance $(X_L = 2\pi fL = 0)$